TCET/EXAM/FRM/IP-03/06 REVISION: B

IN-SEMESTER EXAMINATION-I

F.E./F.T. (Semester-I)

SUBJECT – Mathematics-I

Branch: E&TC, E&CS, IT, MECH, M&ME, AI&ML
Div.: ALL
Duration: 60 Minutes

Date: 28/10/2025
Timing: 12:00 PM to 01:00 PM
Maximum Marks: 20

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions.	Marks	Course Outcomes	Learning Levels
	a.	Verify L.M.V.T for f(x) = $2x^2 - 7x - 1$ on [2, 5] and find the value of c.	2	CO1	R
	b.	Expand 3^x up to first three non-zero terms.	2	CO1	R
	c.	Identify whether the series is convergent or divergent $\sum_{n=1}^{\infty} \frac{n!}{5^n}$.	2	CO1	R
	d.	If $z = (x^3 + y^3 - x^2y)$ then find value of $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$.	2	CO2	R
	e.	If $\varphi = 3z^2x - 3xy - 4x$ then find $\nabla\varphi$ at (1, -1, -1).	2	CO2	R
	f.	Find the value of $(1 + t)^8$ using De'Moivre's theorem.	2	CO3	R
	g.	If $\tanh x = 3$ then find the value of x.	2	CO3	R
Q.2	a.	*Show that* $\lim_{x \to 0} \log_{\sin x}(\tan 2x) = 1$.	5	CO1	A
		OR			
	b.	If $z = f(u, v)$ and $u = \log(x^2 + y^2)$, $v = \frac{y}{x}$ then show that $x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y} = (1 + v^2)\frac{\partial z}{\partial v}$	5	CO2	A

Q.3	a.	Prove that $\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^3}{192} + \ldots$	5	CO1	A
		OR			
	b.	Prove that $\dfrac{1}{1-\dfrac{1}{1-\dfrac{1}{1+\sinh^2 x}}} = -\sinh^2 x.$	5	CO3	A